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 Studierichting: Informatica
 Oefeningenreeks 2E nr. 4.7

$$\begin{aligned}
 & \lim_{x \rightarrow \infty} \left(\frac{ax+b}{ax+c} \right)^x \\
 &= \lim_{x \rightarrow \infty} \left(\frac{(ax+c) + (b-c)}{ax+c} \right)^x \\
 &= \lim_{x \rightarrow \infty} \left(1 + \frac{b-c}{ax+c} \right)^x \\
 & \text{Stel } \frac{1}{u} = \frac{b-c}{ax+c} \Rightarrow u = \frac{ax+c}{b-c} \Rightarrow x = \frac{u(b-c)-c}{a} \\
 &= \lim_{u \rightarrow \infty} \left(1 + \frac{1}{u} \right)^{\frac{u(b-c)-c}{a}} \\
 &= \lim_{u \rightarrow \infty} \left(1 + \frac{1}{u} \right)^{\frac{u(b-c)}{a} - \frac{c}{a}} \\
 &= \lim_{u \rightarrow \infty} \left(1 + \frac{1}{u} \right)^{\frac{u(b-c)}{a}} \cdot \left(1 + \frac{1}{u} \right)^{-\frac{c}{a}} \\
 &= \lim_{u \rightarrow \infty} \left(\left(1 + \frac{1}{u} \right)^u \right)^{\frac{b-c}{a}} \cdot \lim_{u \rightarrow \infty} \left(1 + \frac{1}{u} \right)^{-\frac{c}{a}} \\
 &= e^{\frac{b-c}{a}} \cdot 1 \\
 &= e^{\frac{b-c}{a}}
 \end{aligned}$$

References